# Four Operations for Physics Discovery, and the Guard They All Need

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A breakthrough in physics is rarely a better fit to the data nowadays. More often it is a decision that the data was about something else: Lorentz and Einstein predicted the same numbers, and Einstein's theory dropped an aether velocity no measurement could reach. Automated discovery has no operation for that move. Annotating 2,327 scientific prizes by what the contribution did, we find that among physics prizes **65% changed a description**, while the rung machine learning implements covers **42 of 542** depth labels, one in thirteen; of fourteen automated-discovery systems placed on the same ladder, none performs the move. We build four operations that do: *reinterpretation*, removing structure no measurement can see; *coarse-graining*, finding the level at which a description closes; *transfer*, carrying a solved structure into a new domain; and *positing*, naming an entity that makes a violated conservation law balance. Each works—on a lattice gauge system the first finds five invariances it was told nothing about, separates one redundancy from four symmetries, and emits four conserved charges good to $10^{-10}$. But the result that survives all four is a warning. **Every operation needed two guards, and in each case the intuitive one was nearly worthless.** Two of three vacuous coarse-grainings *close perfectly*; a domain built without the source's structure passes a structural match five times in six; and removing everything invisible destroys conservation laws. Only the guard that hands the system a claim it can *refute* separates the real cases from the empty ones, and it separates them by up to ten orders of magnitude. We then bound the result: on a real controversy the criterion is silent, and four of eight Standard Model concepts were bought by a theorem before any experiment, which no probe-based method can reach. Data, code and negative results are released together at https://real-project-aether.github.io/project-aether/.

## I. PHYSICS DISCOVERY IS MOSTLY REFRAMING

From the outside, science advances by measuring better and fitting better. From the inside, the decisive step is usually something else: someone changes what the measurements are taken to be about, and the old description loses a part it never needed.

In 1905 two theories fitted every optical and electrodynamic measurement equally well. Lorentz kept absolute space and added a contraction hypothesis and a local-time auxiliary, so that the aether became exactly undetectable [1]. Einstein removed the aether [2]. Nothing in the data chose between them: the choice was about which theory carried structure nothing could ever observe.

The distinction is old. Kuhn separated normal science from revolution [3], and Lakatos called an auxiliary that rescues a theory without predicting anything new *ad hoc* [4]. Both verdicts are delivered in retrospect and in prose. What is missing is one that can be computed while the outcome is still open—and an operation that performs the move rather than recognising it afterwards. Sections IV and VI build four such operations and report what building them showed: each needed a second guard, and the first one a designer reaches for accepts answers that are empty.

*a. How much of physics is this?* We annotated a corpus of scientific prizes by *what the contribution did* rather than by how it was cited. Existing prize datasets are bibliometric—who won, what they published, how it was cited [5]—which cannot say whether an advance reframed something or measured something new. A contribution changes one of three things: the **facts** you hold, the **instruments** you have, or the **description** you use. Only the third has a depth, and Fig. 1 gives that ladder. Its six rungs, in order, are **regression** (L0), solving inside the description you already have; **reinterpretation** (L1), the same predictions carried with less structure; **retyping** (L2), changing the level or the quantity the description is written in; **transfer** (L3), carrying one field's language into another; **new object** (L4), adding a kind of thing the language did not have; and **new move** (L5), changing what counts as a move at all. We use these names throughout, and on the project site.

The release holds **2,327 award records** across **23 prizes**, 1731–2026, of which **2,221** carry the awarding body's own citation and **1,547** are annotated. Restricting to the ten physics prizes—the physics Nobel, Dirac, Boltzmann, Onsager, Wolf, Heineman, Kavli, Sakurai, Lorentz and Breakthrough—gives 532 events, and the contrast with science at large is why this paper is about physics:

| | all fields | **physics** |
|---|---|---|
| changed a description | 45% | **65%** |
| changed no description at all | 53% | **33%** |

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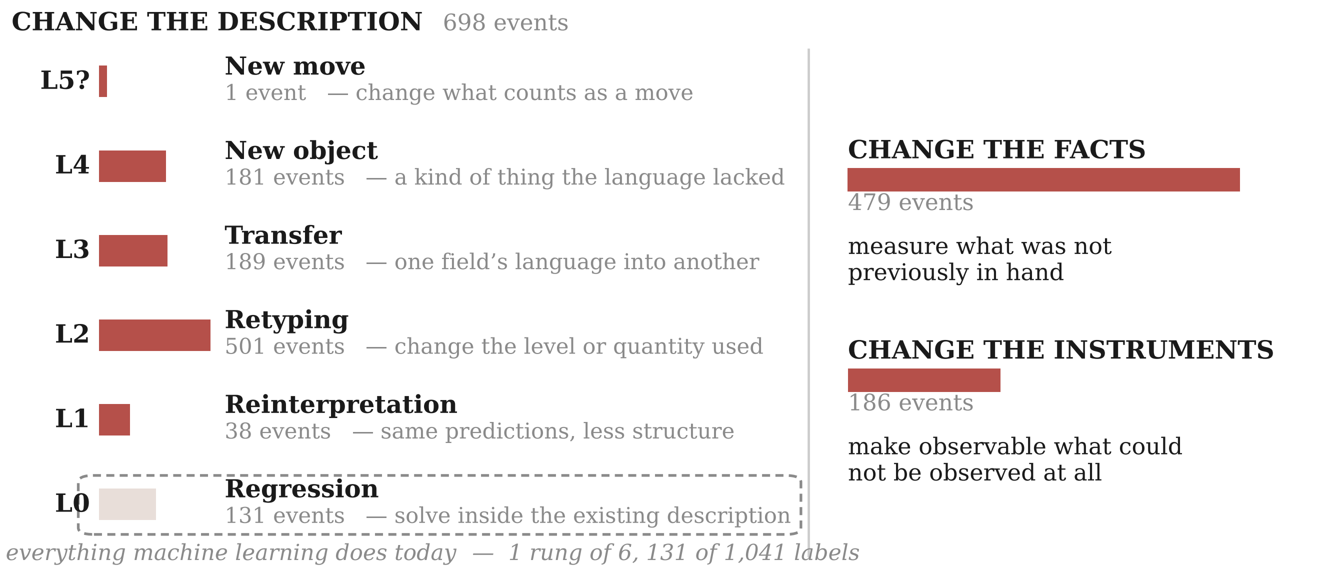


FIG. 1. What a discovering system has to be able to do. A move changes the facts you hold, the instruments you have, or the description you use; a change to the description has a depth. Counts are annotated prize events across all fields. Ordinary machine learning implements the bottom rung.

Across science most awarded work is not a change of description at all, and a framework built only around reframing would miss half its subject. In physics that inverts: two-thirds of awarded work changes the description, so the ladder is not a side issue but the main road. On it, **L0—work inside a fixed description—carries 42 of the 542 physics depth labels**, one in thirteen. L0 is not the shallow end; a Fields Medal can sit there. It is simply the only rung for which machine learning has an operation.

## II. WHAT AUTOMATED DISCOVERY DOES TODAY

Table I places fourteen systems on the same ladder. We classify by the *operation the system exposes*, not by the significance of its results, and the claim is about published operations rather than about limits in principle.

The shape is consistent. Systems that search harder inside a language stay at L0 however impressive the result; DreamCoder reaches L2 by naming reusable structure; predicate invention aims at L4 and is still open. **Nothing performs L1.** No system takes a theory that already fits and returns a theory that fits identically while carrying less.

## III. THE GAP, AND WHY A LANGUAGE MODEL DOES NOT CLOSE IT

A language model can recount the Lorentz–Einstein episode in detail, so it is fair to ask whether the operation is already available in a system that has read the histories. We put five episodes—three historical, two constructed controls—to a 20B-parameter open-weight model [20] in three conditions. The historical three are supernova distance moduli [21] for the acceleration reported in 1998–99 [22, 23] against the grey-dust patch offered for the same dimming [24]; relativistic mass against Lorentz's aether theory; and quasicrystal diffraction [25] against Pauling's multiple twinning [26]. The conditions are **named**, with real theories and domain words; **blind**, every identifier alpha-renamed but fit statistics retained; and **bare**, renamed with statistics withheld.

Table II gives the result. Accuracy is far above chance when the model can recognise the episode ($p = 5\times10^{-9}$) and *indistinguishable from chance* when it cannot ($p =$ 0.9 blind, 0.7 bare; named against blind, paired $p =$ 0.03). The model is recalling these episodes, not judging them—renaming alone is already known to move accuracy on grade-school arithmetic [27], and the blind condition is that probe applied to discovery. One case survives anonymisation: relativity stays at 100% because $(1 - v^2)^{-1/2}$ is recognisable from its algebra whatever the identifiers are called. Handing over the statistics does not rescue the others, so the failure is not a missing-information problem.

## IV. FOUR OPERATIONS, EACH WITH TWO GUARDS

If no system performs these moves, the first question is whether they can be written down as something a machine runs at all. A verdict available only in hindsight and in

TABLE I. Automated discovery systems placed on the ladder of Fig. 1. Placement is by the operation each system exposes. Everything clusters on L0; library learning reaches L2; predicate invention aims at L4 and remains open. **No system implements L1.**

| system | what it does | rung | why not higher |
|---|---|---|---|
| BACON [6] | rediscovers laws from tabulated data | L0 | searches within a fixed term language |
| Eureqa [7] | symbolic regression on motion data | L0 | the operator set is given in advance |
| AI Feynman [8] | physics-informed symbolic regression | L0 | recovers known forms; invents no term |
| PySR [9] | evolutionary symbolic regression | L0 | fixed primitive set |
| LaSR [10] | symbolic regression with a learned library | L0/L2 | abstractions are shortcuts, not retypings |
| DreamCoder [11] | wake–sleep library learning | **L2** | names reusable structure; no theory to reinterpret |
| FunSearch [12] | LLM program search against an evaluator | L0 | searches programs in a fixed language |
| AlphaGeometry [13] | olympiad geometry by synthesis and search | L0 | derives inside a fixed axiom set |
| Robot Scientist [14] | forms and tests hypotheses with a robot | L0 + facts | chooses experiments; the ontology is fixed |
| Coscientist [15] | plans and runs chemistry autonomously | instruments | extends reach; no description changes |
| AI Scientist [16] | end-to-end machine-learning papers | L0 | searches a fixed hypothesis space |
| AlphaFold [17] | protein structure prediction | tractability | makes computable; does not reframe |
| Cascade-Correlation [18] | grows hidden units as needed | L0 | capacity, not concepts |
| predicate invention [19] | invents relations in logic programs | **L4***| *open after three decades of ILP |

prose cannot be automated. Each operation below turns a move the corpus records into a computation, and each needs *two* conditions to accept a candidate. Section VI is about why.

*a. L1, reinterpretation.* "Carries structure no measurement can see" is not a metaphor. For a theory with parameters $p$ predicting $f(x, p)$, the directions along which no prediction moves are the null space of the Jacobian $J_{ij} = \partial f_i/\partial p_j$—structural non-identifiability [28], read as a property of a theory rather than a nuisance in a fit. Exact degeneracy is the wrong test for real data (Lorentz's aether was not unobservable in principle, but to within the precision of the day), so we scale column $j$ by $|p_j|$ and row $i$ by $1/\sigma_i$ and ask whether a direction moves by an order-unity amount without shifting any prediction past its error bar.

Removing every invisible direction is wrong, and wrong in a way that matters. A *global* phase rotation is invisible to every probe exactly as a gauge transformation is; removing both does not discover charge conservation, it **erases** the conserved quantity. The second guard is a question about locality:

| | | |
|---|---|---|
| invariant *point by point* | redundancy: quotient it | [L1] |
| invariant only *uniformly* | symmetry: keep it, read off a conserved charge | [L4] |

One test, two moves, and the second is the more productive: *a distinction no probe can see, but only when moved everywhere at once, is a conservation law.* Our implementation refuses to quotient at all when no locality test is available, rather than guess.

*b. L2, coarse-graining.* L1 asks what *parameter* detail predictions do not need; this asks what *state* detail the future does not need. A reduction $c$ maps fine states to coarse ones, and *closes* if the coarse variables predict their own future without the fine state. We test closure twice on purpose: with a coarse law *learned* from coarse readings alone, which is the situation a discovering system is in, and one *derived* by projecting the known fine law [29]. If the derived law closes and the learned one does not, the failure was our fitter and not the physics.

The second guard is that the reduction must *pay*: the coarse state must still carry the fine observables. Without it the operation is empty, because a reduction that maps every state to zero closes perfectly.

*c. L3, transfer.* Two systems, each with states, probes and dynamics. If a map carries one domain's states into another's while preserving what the probes see, then what the first domain knows becomes a claim about the second. The first guard is that structural match. The second is that the transfer must pay: what crosses has to be a claim the target did not have and its own dynamics can refute—here a conserved quantity, which the target either conserves or does not.

*d. L4, positing an entity.* This consumes what L1's symmetry branch produces. That branch *finds* conserved charges; this one *uses* one. When a found charge fails to balance across an observed process, the missing amount is a description of something not yet named:

TABLE II. Accuracy at identifying which of two candidates is the reinterpretation. Twelve repeats per cell, presentation order randomised; the two controls are averaged. Chance is 50%.

| episode | named | blind | bare |
|---|---|---|---|
| dark energy | 67% | 42% | 50% |
| relativity | 100% | 100% | 100% |
| quasicrystals | 100% | 50% | 42% |
| two controls | 84% | 25% | 38% |
| pooled, 60 trials | **87%** | 48% | 53% |

writing $\Delta Q$ for charge in minus charge out,

$$\Delta Q = \begin{cases} 0 & \text{nothing to posit} \\ Q \neq 0 & \text{posit a carrier of } Q \end{cases}$$

The second guard is the noise scale rather than significance: the imbalance must exceed the systematic uncertainty, not merely many standard errors from zero.

*e. What proposes.* Every operation above can only *reject*. Something must propose, and a null-space search finds only what its basis contains. Section III says what a language model is for here: it recalls fluently and judges at chance. We use it *only* to generate, never to decide.

## V. RESULTS

*a. Controls first.* Twenty-one checks run on cases whose answers we put in ourselves: the mechanism must find hidden slack and remove it *without moving any prediction*, leave an honest theory alone, and **reject** a proposed term that buys coverage—the fraction of observations the fitted theory reproduces inside $2\sigma$—while moving no prediction. All twenty-one pass. This establishes that the implementation does what it claims, not that the criteria track real theory choice.

*b. L1: discovering an unknown symmetry.* On a periodic $6 \times 6$ lattice carrying a complex two-flavour matter field and U(1) links, probed only by gauge-invariant, flavour-blind quantities, the mechanism is handed eight candidate generators and told nothing about which are real invariances. The null space of $\partial(\text{probes})/\partial c$ under uniform application has rank 3, so **five invariant directions are found**; a link phase, a rescale and a single-flavour shift are rejected. A second null-space computation *inside* the first, with the coefficient now varying from site to site, isolates **one redundancy** (local response $2.6\times10^{-5}$) from **four symmetries** ($5.1$–$6.6\times10^{1}$). The order matters: at uniform order the gauge transformation and the global phase act identically, so the first null space mixes them and only the local test breaks the degeneracy.

| branch | relative drift |
|---|---|
| four symmetry charges | $1.8\times10^{-11}$–$7.6\times10^{-10}$ |
| redundancy, would-be charge | $8.6\times10^{-1}$ |
| energy, same trajectory | $1.3\times10^{-10}$ |

The charges are conserved at the trajectory's own precision and the redundancy's would-be charge is nine orders of magnitude worse. **Without the local test, all five directions are unobservable and all five would be quotiented—destroying four conservation laws to remove one redundancy.**

*c. L2: finding the level at which a description closes.* On a system with four slow variables coupled to eight fast ones, with the separation set by one knob, the operation accepts at $62.9\times$ and $25.2\times$ separation (closure error 0.062 and 0.156, retention 0.95 and 0.91) and refuses below, placing the sensitivity floor between $12\times$ and $25\times$. On the lattice above it accepts nothing at any level: closure error falls monotonically to zero only at the identity, and learned and derived agree to three decimals (1.626 against 1.628), so that negative is the physics rather than our fitting. A nonlinear field theory whose spectrum spans $2\times$ has no separation to exploit, which is consistent with the measured floor.

*d. L3: carrying a structure into a new domain.* A source domain with a known conserved charge, and a larger target whose containing of that structure is what must be discovered. Across six seeds the real transfers are accepted 6/6 (charge drift at most $8.4\times10^{-11}$) and matched controls, built without the shared structure, are accepted 0/6.

*e. L4: positing what closes a ledger.* Three species of charge $+2, +1, +1$ with the third invisible to every probe; charge is conserved by construction. When a product lands in the invisible species the operation posits a carrier of $+1.192$ per event at $56\sigma$, against a withheld truth of 1.200—under 1% error. It stays silent when the books balance ($-0.011$, $-1.0\sigma$) and when the imbalance is noise ($+0.029$, $2.5\sigma$). It also stays silent at $16\sigma$ and $30\sigma$ when the imbalance is below the systematic uncertainty, which is the correct answer: positing a particle on a 5% effect with 2% calibration is how false discoveries are made.

*f. The proposer, and its filter.* On symbolic worlds the model proposes candidate concepts and the acceptance rule decides. Across three runs, a world whose residual hides $0.4 \sin 3x$ yielded eight proposals of which two to five were accepted; a control whose residual is pure noise yielded eight proposals of which **none** was accepted, in every run. Offered the true concept directly the filter is decisive—$\sin 3x$ reaches 97% coverage against 68% for $\sin 2.9x$ and 20% for wrong terms—so the limiting component is the proposer, not the test: the model offered $\sin x$, $\sin 2x$, $x^3$ and $\sin \pi x$, never $\sin 3x$, though the residual is a clean sine.

*g. Where the criterion goes silent.* We applied L1 to a live controversy with no known answer: the rotation curve of NGC 3198 from SPARC [30], fitted by a dark-matter halo and by MOND [31]. All three descriptions carry *zero* unobservable directions; the halo model's softest direction still moves the curve $6.4\sigma$ under a 100% excursion, unsurprising since SPARC's $3.6\,\mu$m photometry was assembled to break exactly that degeneracy. **The criterion is silent, and that is the result.** The Lorentz–Einstein signature needs two descriptions that *agree* on the data; these disagree at $\chi^2/N$ of 1.22 against 9.21, and where descriptions disagree ordinary model comparison settles it.

## VI. THE FINDING: THE INTUITIVE GUARD IS NEARLY WORTHLESS

Each operation was designed with an obvious acceptance test, and in each case that test turned out to accept vacuous answers. The pattern was not designed in; it appeared three times independently, and it is the most transferable thing here.

The clearest case is L2. A reduction that maps every state to zero closes *perfectly*: its coarse dynamics is trivially autonomous. So does a reduction onto the *fast* modes, because fast modes decay to zero—a description of nothing, evolving correctly. Both satisfy the guard a designer would write first. Only retention refuses them, and it refuses them at the floor.

L3 shows the same thing with a control that is not degenerate at all. A domain built without the source's structure passes the structural match in **five of six** seeds: fit a map with enough freedom and two domains can be made to look alike. The transferred-claim test separates real from control by ten orders of magnitude ($1.5\times10^{10}$ between the worst real case and the best control) because it is not a similarity score—it is a prediction the target can break.

L1's version is the sharpest, because there the naive guard does not merely accept something empty, it destroys something real: quotienting every invisible direction removes four conservation laws to eliminate one redundancy.

We draw two conclusions. First, for anyone building such systems: *a structural match is not evidence*. Approaches that find an analogy, a symmetry or a reduction and stop there are measuring almost nothing, and the literature on analogy-driven discovery should be read with that in mind. Second, for us: a guard is only trustworthy if it carries an invariant that fails loudly. Our own retention guard was wrong three times—fitted on features that could not represent the observables, then overfitted at a perfect in-sample $R^2$ on 62 samples, then dominated by scale spread—and each error was caught by an invariant, never by a plausible number. The version we kept is fit-free and must score exactly 1.000 when the reduction keeps everything. It does.

TABLE III. The first guard is the one a designer reaches for. In each case it accepts answers that are empty, and only the second—which hands the system a claim it can refute—separates them.

| | the intuitive guard | how it fails |
|---|---|---|
| L1 | remove what no probe sees | deletes conservation laws |
| L2 | the coarse variables close | 2 of 3 empty cases close |
| L3 | the structures match | control passes 5 of 6 |
| L4 | the imbalance is significant | fires below systematics |

## VII. WHAT PROBE-BASED DISCOVERY CANNOT REACH

The mechanism above buys concepts with measurements. Tested against eight concept-forming episodes in the Standard Model it accounts for **two**: the neutrino, forced by a continuous spectrum where a two-body decay demands a line, and gauge redundancy. The other six do not fail for want of cleverness. Four of them contain *no probe at all*:

- $\Delta^{++}$ violated Fermi statistics. Colour was invented so the state would be *allowed*.
- Flavour-changing neutral currents were too suppressed. Charm was invented so the excess would *cancel*.
- Gauge anomalies had to sum to zero, which required *complete* generations.
- $WW$ scattering had to stay unitary, which put the Higgs below about a TeV—and told experimenters what to build.

Every one predicted a real particle. Every one was paid for by a **theorem first**, with the experiment arriving years or decades later. A framework whose founding commitment is that a concept is paid for by an experiment has these exactly backwards. The prerequisite is not more data or a better search: *you can only have theorems if there is a formal theory to have them about*, and no discovery system we know of carries one. That, rather than the difficulty of reinterpretation, is what bounds automated physics discovery today.

The weaker version of this route is already refuted here. Proposing concepts from structural defects in a description—dangling structure, role collisions—was tested on three substrates and lost on all three. The difference between that and the Standard Model is grade, not kind: heuristics say a defect is a hint, theorems say a state is forbidden or it is not.

*a. And one rung we did not attempt.* L5, changing the repertoire of moves itself, has one event in 1,547 and the annotator flagged even that one uncertain. Across 163 mathematics prizes—where inventing a *method* rather than proving a *theorem* is routine—the scheme records exactly one. More to the point, every operation above has a measurable trigger: a null space, a closure failure, a structural match, an imbalance. L5's would be "the existing repertoire is insufficient", and a repertoire cannot measure its own insufficiency from inside. We did not build it, and record here that this is a statement about the problem rather than about our effort.

## VIII. LIMITATIONS

*a. One annotator, and it is a model.* Every count in Fig. 1 and in Sec. II is one annotator's judgement, and that annotator is a large language model reading each official citation under human direction. These are not the judgements of human domain experts; there is exactly one annotator, so no inter-annotator agreement statistic exists and we do not report a proxy for one. The boundary that matters most is between recording facts and changing a description, and it is a judgement call. Independent re-annotation is the single most useful contribution a reader could make.

*b. The criterion was revised against known answers.* An earlier form of this test was modified eight times, each after seeing a verdict we knew to be wrong; a description-length score [32] was among the versions that failed. Five episodes whose outcomes we already knew establish internal consistency, not predictive power, which is why the results above rest instead on a lattice system with an independently known answer and on a controversy with no known answer at all.

*c. The proposers are narrow.* Coarse-graining searches only *linear* reductions, taken from the linearised dynamics; Kuramoto's phase reduction, a canonical instance of the move, is not a linear projection and would not be found. Transfer likewise searches linear maps, while the corpus's real transfers—statistical mechanics into optimisation, quantum fluctuations onto cosmological scales—are not linear recodings.

*d. Two of the four operations are tested only on constructed systems.* Coarse-graining and transfer were run on systems we built, one with a coarse level and one without, one containing a shared structure and one not. Neither has been run against a physical system independently known to have an effective description.

*e. L4 assumes the conservation law.* It reads the residual off a law it is handed. The hard half of Pauli's move was *deciding to trust conservation over the measurements*—Bohr was prepared to abandon energy conservation instead—and nothing here makes that choice. What is posited is also a quantity of charge, not an entity with mass, spin and statistics.

*f. The acceptance rule does not discriminate.* "Coverage strictly improves" admitted $\sin x$ and $x^3$ at 20–22% against a base of 18%. It ranks correctly but needs a margin, not a strict inequality.

*g. The scheme is not symmetric.* The facts column distinguishes facts newly in hand from reach newly extended, but encodes that as two categories rather than as a ladder. A scheme that graded every column would be cleaner; we did not build one.

*h. L2 is a residual category, not a move.* Of its 501 events, 34% carry a coarse-graining operator and only 7% carry the merge-or-split operator by which we first defined the rung; a further 10% and 7% carry operators belonging to L4 and L3. L2 is where events land when they change the description but are not cleanly L1, L3 or L4. We have renamed it after its dominant sub-move, changing the level or quantity a description is written in, but the honest reading is that this rung should be cut into two or three.

*i. Hindsight and selection.* Prizes are awarded with decades of hindsight, so the corpus samples work that turned out to matter, and citations justify a decision already taken.

*j. Availability.* Everything is at `https://real-project-aether.github.io/project-aether/` — the corpus, the mechanism, both negative results, and a script that recomputes every figure in this paper from the shipped data and fails loudly if one does not reproduce. Our annotations are CC BY 4.0; the citation text remains the awarding bodies'.


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